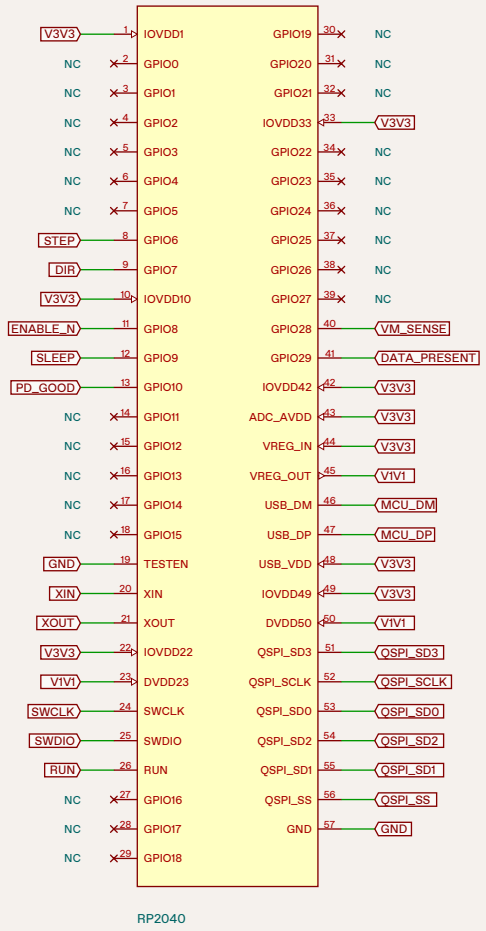
NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 1/13 - U_MCU

U_MCU



RP2040

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

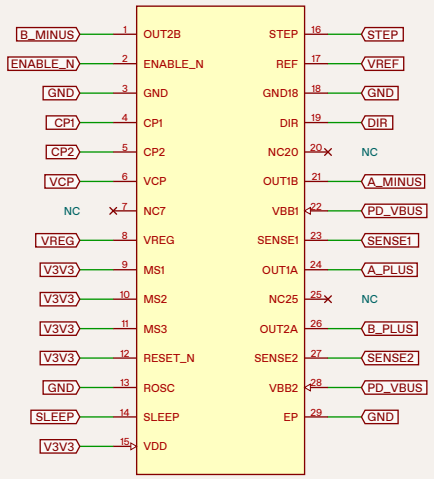
U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 2/13 - U_DRV

U_DRV



A4988SETTR-T

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

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Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

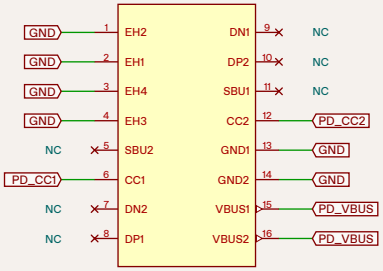
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

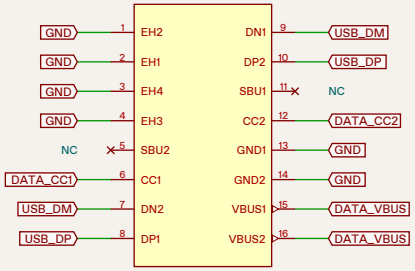
SHEET 3/13 - J_PD, J_DATA, U_PD, R_PD, C_PD, R_PG

J_PD



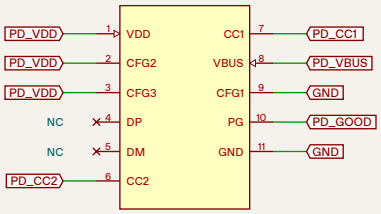
TYPE-C-31-M-12

J_DATA



TYPE-C-31-M-12

U_PD



CH224K







Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

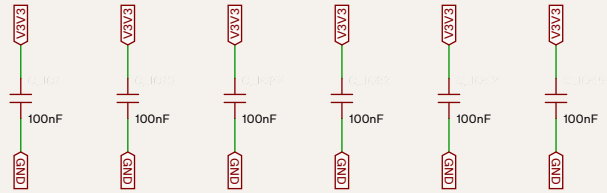
Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 4/13 - C_IO1, C_IO10, C_IO22, C_IO33, C_IO42, C_IO49



COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

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Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

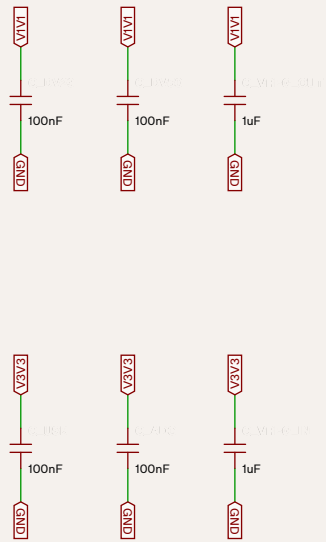
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 5/13 - C_DV23, C_DV50, C_USB, C_ADC, C_VREG_IN, C_VREG_OUT



Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support
oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

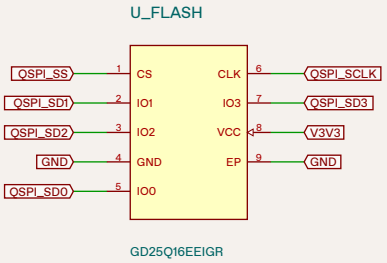
Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

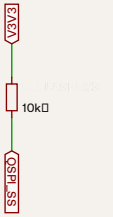
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

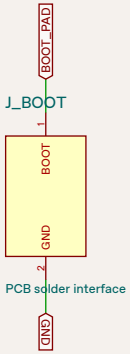
SHEET 6/13 - U_FLASH, C_FLASH, R_FLASH_CS, R_BOOT, J_BOOT, Y_MCU

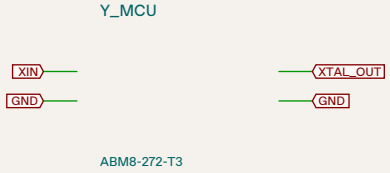












COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

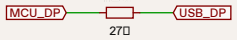
SHEET 7/13 - R_XOUT, C_XIN, C_XOUT, R_RUN, R_DM, R_DP













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PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 8/13 - R_CC1, R_CC2, R_USB_SENSE_H, R_USB_SENSE_L, R_VM_H, R_VM_L



Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
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Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
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U_BUCK - AP63203WU-7

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Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

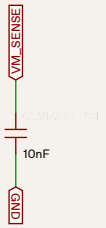
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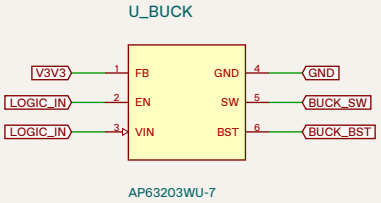
U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

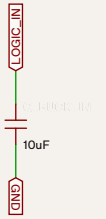
SHEET 9/13 - C_VM_SENSE, U_BUCK, L_BUCK, C_BOOTSTRAP, C_BUCK_IN, C_BUCK_OUT1













COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
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U_ESD - USBLC6-2SC6

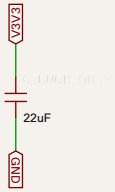
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 10/13 - C_BUCK_OUT2, D_PD, D_DATA, C_CP, C_VCP, C_VREG













COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

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2 MB external QSPI flash stores firmware.
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U_ESD - USBLC6-2SC6

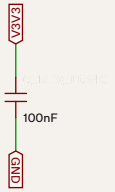
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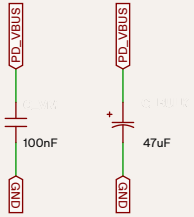
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NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 11/13 - C_DRV_LOGIC, C_VM, C_BULK, R_SA, R_SB, R_REF_H











COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

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12 MHz crystal is the USB clock reference.
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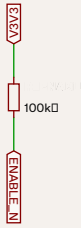
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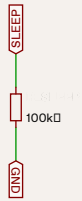
NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 12/13 - R_REF_L, C_REF, R_ENABLE, R_SLEEP, J_MOTOR, J_DEBUG

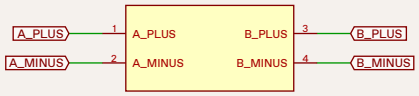






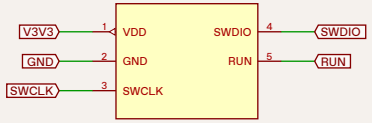


J_MOTOR



B4B-PH-K-S(LF)(SN)

J_DEBUG



PCB solder interface

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
Nominal phase-current limit is 0.351 A.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

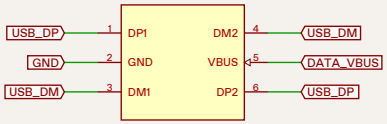
Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

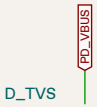
SHEET 13/13 - U_ESD, D_TVS

U_ESD



USBLC6-2SC6

D_TVS



SMF18A



COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support oscillation and limit crystal drive.

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